



Quadriceps mitochondrial DNA quantity, quality, and gene expression after 2 years of calorie restriction: exploratory results from the CALERIE trial

Jayanta K. Das · Yu Zhang · Leanne M. Redman · Corby K. Martin · Sai Krupa Das · Susan B. Racette · Allen Herbst · Nianjun Liu · David B. Allison · Debbie McKenzie · Judd M. Aiken · Steve Horvath · Julián Candia · Nirad Banskota · Luigi Ferrucci · William E. Kraus · Kim M. Huffman · Jonathan Wanagat 

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Abstract The Comprehensive Assessment of Long-term Effects of Reducing Intake of Energy (CALERIE)TM trial was a randomized, 2-year controlled trial of caloric restriction (CR) versus an ad libitum (AL) control condition in nonobese humans. We performed exploratory analyses of muscle mitochondrial DNA (mtDNA) integrity, one element of mitochondrial quality. Our aims were to assess the feasibility of this approach, explore associations to inform future hypotheses, estimate effect sizes and statistical power for subsequent studies,

and contribute additional data to the CALERIE database. We used droplet digital PCR to quantitate the copy numbers of nuclear DNA, mtDNA, and mtDNA deletion mutations in remnant total DNA samples extracted from quadriceps biopsies at baseline ($n=93$), 12 months ($n=44$), and 24 months ($n=31$). MtDNA copy number and mutation frequency were correlated with existing gene expression data from the same muscle biopsies. MtDNA copy number was lower in females ($p=0.0005$) and declined over time ($p=0.0001$), with no statistically significant differences observed for CR versus AL ($p=0.2898$) or age across both groups ($p=0.4644$). Baseline copy number correlated positively with baseline physiological measures including fat-free mass ($r=0.43$, $p=2e-05$), self-reported energy intake ($r=0.34$, $p=0.00077$), resting metabolic rate ($r=0.42$,

Jayanta K. Das and Yu Zhang contributed equally to this work.

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J. K. Das · J. Candia · L. Ferrucci
Longitudinal Studies Section, Translational Gerontology Branch, National Institute on Aging, National Institutes of Health, Baltimore, MD, USA

Y. Zhang · N. Liu · D. B. Allison
Department of Epidemiology and Biostatistics, Indiana University, Bloomington, USA

L. M. Redman · C. K. Martin
Pennington Biomedical Research Center, Baton Rouge, LA, USA

S. K. Das
Jean Mayer USDA Human Nutrition Research Center on Aging at Tufts University, Boston, MA, USA

S. B. Racette
College of Health Solutions, Arizona State University, Phoenix, AZ, USA

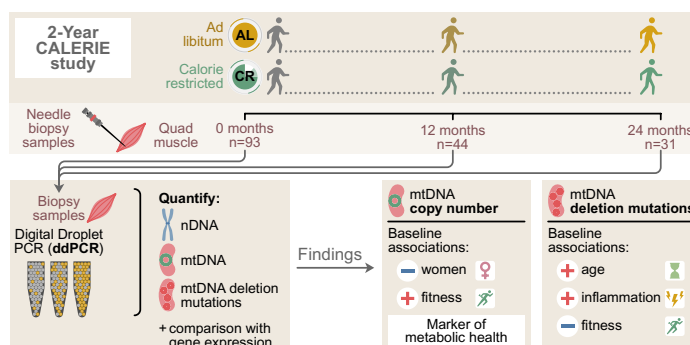
A. Herbst · J. M. Aiken
Department of Agricultural, Food, and Nutritional Science, University of Alberta, Edmonton, Alberta T6G 2R3, Canada

D. B. Allison
USDA –ARS Children’s Nutrition Research Center, Baylor College of Medicine and Texas Children’s Hospital, Houston, USA

$p=0.00261$), total energy expenditure ($r=0.31$, $p=0.00261$), and VO_2max ($r=0.41$, $p=1\text{e}-04$), and with gene expression related to mitochondrial function, while showing negative correlations with nuclear genome maintenance, RNA splicing, and ribosomes. MtDNA mutation frequency increased with age ($p=0.0041$) and showed a weak negative correlation with VO_2max ($r=-0.23$, $p=3\text{e}-05$). MtDNA mutation frequency was not statistically different with CR. Mutation frequency was positively

correlated with gene expression changes in inflammation and negatively correlated with protein translation. We observed that, as expected, mtDNA copy number declined over time, was lower in females, and correlated with metabolic measures and mitochondrial gene expression, suggesting its potential utility as a marker of metabolic health. These findings contribute to understanding mitochondrial adaptations to CR and highlight potential benefits of CR interventions.

Graphical Abstract



Keywords Calorie restriction · Human · Mitochondria · DNA · Mutation · Gene expression

D. McKenzie
Department of Biological Sciences, University of Alberta,
Edmonton, Canada

S. Horvath
Human Genetics, David Geffen School of Medicine,
UCLA, Los Angeles, USA

S. Horvath
Altos Labs Limited, Cambridge, UK

N. Banskota
Computational Biology and Genomics Core, National
Institute on Aging, National Institutes of Health,
Baltimore, MD, USA

W. E. Kraus · K. M. Huffman
Duke Molecular Physiology Institute, Department
of Medicine, Duke University School of Medicine,
Durham, NC, USA

J. Wanagat (✉)
Division of Geriatrics, Department of Medicine,
University of California, Los Angeles, 650 Charles E.
Young Drive South, RM 34-115, Los Angeles, CA 90095,
USA
e-mail: jwanagat@mednet.ucla.edu

J. Wanagat
Veterans Administration Greater Los Angeles Healthcare
System, Los Angeles, USA

Introduction

Aging and age-related diseases represent a major global burden, affecting quality of life and increasing healthcare costs [1]. Mitochondrial dysfunction is a long-recognized hallmark of aging and may contribute to the progression of age-associated metabolic and degenerative diseases, as well as cancer and neurological disorders [2]. Mitochondria are essential for cellular energy production and play critical roles in cell signaling, differentiation, and programmed cell death. Alterations in mitochondrial DNA (mtDNA) are closely correlated to aging and contribute to its phenotypic manifestations [3, 4]. Understanding the mechanisms underlying mitochondrial decline may inform interventions aimed at delaying aging and mitigating age-related diseases.

Calorie restriction (CR), defined as a reduction in caloric intake without malnutrition, has been extensively studied for its potential to extend lifespan and delay the onset of age-related disease across multiple species [5, 6]. CR induces numerous physiological changes, including improved mitochondrial function and increased mitochondrial biogenesis [7, 8]. These adaptations may enhance metabolic efficiency, reduce oxidative stress, and support cellular repair processes [8–10].

The mitochondrial genome encodes essential components of the respiratory chain, which are required for mitochondrial energy transduction. Maintaining mtDNA integrity is critical for mitochondrial and cellular function and is thought to support healthy aging. In humans, skeletal muscle mtDNA copy number decreases approximately linearly with age, while the frequency of mtDNA deletion mutations increases approximately exponentially [11–13]. In muscle tissue, accumulated mtDNA deletion mutations disrupt respiration and electron transport and trigger cell death [14, 15]. While studies in model organisms suggest that CR can modify mtDNA copy number and reduce mutation burden through unknown mechanisms [9, 15, 16], data in humans remain limited and sometimes conflicting. For example, in a 6-month trial, CR increased skeletal muscle mtDNA copy number by 35% and was associated with increased expression of nuclear-encoded mitochondrial genes, though mitochondrial enzyme activities remained unchanged [9]. In contrast, a 12-month study found that CR decreased skeletal muscle mtDNA copy number by 15% with time but not with CR [8]. Both

studies used quantitative PCR to measure mtDNA copy number but did not report assay validation details, such as standard curves or PCR efficiency, and neither examined the effects of CR on mtDNA mutations.

In this exploratory analysis, we examined skeletal muscle biopsy samples from participants in the 2-year, multicenter, randomized controlled Comprehensive Assessment of Long-term Effects of Reducing Intake of Energy (CALERIE) trial to assess mtDNA copy number, nuclear copy number, and mtDNA deletion mutation frequency using digital PCR. This method provides absolute quantification, is independent of PCR efficiency, and does not require standard curves [17]. We employed a linear mixed-effects model (LME) to assess the longitudinal changes in mtDNA copy number and mutation frequency over time within and between AL and CR groups. At baseline, we evaluated associations between mtDNA measures, muscle-related physiological outcomes, and skeletal muscle gene expression profiles. These findings may inform future research into the role of CR in modulating mitochondrial genome integrity and function in humans.

Materials and methods

CALERIE study characteristics

The CALERIE trial compared a 2-year 25% CR diet to an ad libitum (AL) control condition among adults without obesity [18, 19]. Details of the CALERIE muscle biopsy study methodology and participants were described previously [20]. Briefly, a subset of CALERIE study AL and CR participants consented to skeletal muscle biopsy (Table 1). Biopsies were planned at baseline, 12, and 24 months. Our mitochondrial DNA and transcriptional analyses were not prespecified in the pre-registered clinical trial (NCT00427193) but were secondary analyses of remnant DNA samples after DNA methylation studies had been completed. The CALERIE trial was carried out from January 2007 to November 2012 at three locations: Pennington Biomedical Research Center in Baton Rouge, LA; Washington University School of Medicine in St. Louis, MO; and Tufts University in Boston, MA. The Duke Clinical Research Institute, NC, coordinated the study. The protocol received

Table 1 Demographic and anthropometric characteristics of participants with muscle biopsies in the CALERIE trial

	Baseline (<i>n</i> = 93)			Time 12 (<i>n</i> = 44)			Time 24 (<i>n</i> = 31)		
	AL (<i>n</i> = 34)	CR (<i>n</i> = 59)	<i>p</i> -value	AL (<i>n</i> = 14)	CR (<i>n</i> = 30)	<i>p</i> -value	AL (<i>n</i> = 11)	CR (<i>n</i> = 20)	<i>p</i> -value
Race			0.771 ^a			0.513 ^a			1 ^a
African American	4 (12)	7 (12)		1 (7)	3 (10)		0 (0)	1 (5)	
Other	3 (9)	3 (5)		1 (7)	0 (0)		0	0	
White	27 (79)	49 (83)		12 (86)	27 (90)		11 (100)	19 (95)	
Sex			0.223 ^b			0.290 ^b			0.640 ^c
Female	20 (59)	42 (71)		7 (50)	20 (67)		6 (55)	14 (70)	
Male	14 (41)	17 (29)		7 (50)	10 (33)		5 (45)	6 (30)	
Age, years	40.1 (1.1)	38.4 (0.96)	0.249	40.9 (1.7)	40.1 (1.23)	0.683	43.5 (2.49)	41.1 (1.54)	0.423
Height, cm	169.4 (1.58)	168.7 (1.08)	0.702	171 (2.54)	169 (1.24)	0.479	173.4 (2.57)	168.5 (1.42)	0.115
Weight, kg at baseline	72.8 (1.72)	72.2 (1.11)	0.745	73.0 (2.72)	72.2 (1.52)	0.783	76 (2.71)	71.2 (1.78)	0.157
BMI at baseline	25.3 (0.31)	25.3 (0.22)	0.925	24.9 (0.41)	25.2 (0.30)	0.514	25.2 (0.52)	25.0 (0.36)	0.734

^aFisher's exact test^bPearson's chi-squared test^cPearson's chi-squared test with Yates' continuity correctionDate for categorical variables (race, sex) are *n* (%); data for continuous variables (age, height, weight, BMI) are mean (SEM)

approval from the institutional review boards at each site and the coordinating center, and participants gave written informed consent.

Total DNA isolation and quality

De-identified human muscle biopsy samples were powdered under liquid nitrogen using a mortar and pestle. Powdered muscle was used for DNA isolation performed by proteinase K and RNase A digestion, phenol/chloroform extraction, and ethanol precipitation, as previously described [11]. Quality control of the total DNA was performed via microvolume spectrophotometer (NanoDrop 2000, Thermo Fisher Scientific; Waltham, MA), Qubit (2.0, Invitrogen; Carlsbad, CA), and microfluidic electrophoresis (TapeStation 2200, Agilent; Santa Clara, CA).

Droplet digital PCR quantification of nDNA, mtDNA, and mtDNA deletion mutations

Droplet digital PCR (ddPCR) was used to quantify copy numbers for nuclear DNA, mtDNA, and mtDNA deletion mutations with specific primer/probe sets and cycling conditions for each as previously described [11]. Samples were diluted to the

manufacturer's recommended target range (1 to 5000 copies per microliter) in 25 µl reactions using BioRad ddPCR Supermix for Probes (BioRad; Hercules, CA). Final primer and probe concentrations were 900 nM and 250 nM, respectively. Reactions were partitioned into droplets using a BioRad QX200 Droplet Generator (BioRad; Hercules, CA) prior to thermocycling. Digital PCR cycling conditions for nDNA and mtDNA copy number included Taq-polymerase activation at 95 °C for 10 min, followed by 40 cycles of denaturation at 94 °C for 30 s and annealing/extension at 60 °C for 2 min. Droplet fluorescence was then read using a BioRad QX200 Droplet Reader. Target copy numbers per microliter were determined using BioRad QuantaSoft Regulatory Edition Software (Version 1.7, BioRad; Hercules, CA). Direct quantitation of mtDNA major arc deletions by ddPCR used cycling conditions of Taq-polymerase activation at 95 °C for 10 min, followed by 60 cycles of denaturation at 94 °C for 15 s, annealing at 55 °C for 1 min, and extension at 72 °C for 6 min.

Statistical analyses of mtDNA measures

Age, height, weight, body mass index (BMI), and mtDNA copy number were shown as means

(standard error of the mean (SEM)). MtDNA deletion mutation frequency was shown with median (interquartile range (IQR)) because of its extremely skewed distribution. Race and sex were summarized using frequency counts and percentages to provide an overview of the sample distribution. For continuous variables that meet the assumptions of normality, the two-sample Student's *t*-test was employed to evaluate group differences. In cases where non-normality was detected, the non-parametric Wilcoxon rank-sum test was utilized. Categorical variables were analyzed using Pearson's chi-squared test with appropriate adjustments for small sample sizes (Yates' continuity correction) or sparse data (Fisher's exact test). A log transformation was applied to the mtDNA deletion mutation frequency prior to the comparison as this frequency is known to increase exponentially with age and a log transformation approximated normality. To assess the change in outcome variable over time within and between groups, we employed a linear mixed-effects model (LME) and model assumptions were checked. This approach accounted for the nested structure of the data, where repeated measurements were collected from individuals within each group. Missing data imputation was not performed because the LME model inherently accounts for data missing completely at random by leveraging all available observations, providing unbiased estimates of fixed effects without requiring imputation [21]. Pearson's correlation coefficients were employed to investigate the magnitude and direction of the linear association between outcome variables and selected muscle-related physiologic measures that included whole-body fat-free mass, self-reported energy intake (KCAL), resting metabolic rate (RMR), total energy expenditure from double-labeled water with individual respiratory quotient (TEERQ), and VO2max as detailed at <https://calerie.duke.edu/protocols-procedures>. Exact *p*-values and the number of tests performed are reported for transparency; multiple testing corrections were not applied. All *p*-values reported are two-sided when applicable.

Regression analyses of gene expression, mtDNA copy number, and mutation frequency

Regression analysis was performed with 88 participants for whom gene expression data were available

(55 CR and 33 AL) [20] as well as mitochondrial DNA copy number and mitochondrial DNA mutation frequency data. Gene expression data were normalized using log2(CPM), as outlined in [20]. MtDNA copy number data were standardized (*z*-score) to place mtDNA copy number on a comparable scale with other predictors and to facilitate interpretation of effect sizes. MtDNA mutation frequency data were log transformed (base10).

For the baseline cross-sectional analysis, we performed linear regression analysis between gene expression and mtDNA copy number and mtDNA mutation frequency, where gene expression is considered the outcome variable, with copy number or mutation frequency serving as the predictor variable. We adjusted for age, sex, race, and technical batch in the regression model (Eq. 1). A significance threshold of $p < 0.05$ (unadjusted, two-tailed) was utilized for determining the statistical significance in the regression analysis for understanding the association between each individual gene and mtDNA copy number and mtDNA mutation frequency. Our analysis focused on 16,450 genes after filtering out lowly expressed genes as reported in [20].

$GeneExp \sim \beta_0 + \beta_1 mtDNA\ copy\ number\ (or\ mtDNA\ mutation\ frequency)$

$+ covariates(age + sex + race + batch) + \epsilon_i$

(1)

In our longitudinal study, we employed a random intercept linear mixed-effects model, ensuring consistency with the outcome, predictors, and adjusted covariates as delineated in our baseline investigation. Our main objective was to discern the contrasting changes between the CR intervention group and the AL control group, achieved through interaction terms such as mtDNA copy number (or mtDNA mutation frequency) * group (CR, AL) * linear time. These interactions shed light on the changes of copy number or mutation frequency over time, thereby aiding in the prediction of gene expression patterns. To conduct the mixed model analysis, we utilized the “lmerTest” package in R (<https://github.com/runehaubo/lmerTestR>). By default, the linear mixed model was fitted using REML, and statistical significance was determined through *t*-tests using Satterthwaite's method. A significance threshold of $p < 0.05$ (unadjusted, two-tailed) was utilized for determining the statistical significance in the linear mixed effect model for understanding the association between each individual gene

and mtDNA copy number and mtDNA mutation frequency focusing on contrasting changes between the CR intervention and AL groups over time.

$$\begin{aligned} \text{GeneExp} \sim & \beta_0 + \beta_1 \text{mtDNA copy number (or mtDNA mutation frequency)} \\ & * \text{group} * \text{timepoint} + (1|\text{subject}) \\ & + \text{covariates}(\text{age} + \text{sex} + \text{race} + \text{batch}) + \epsilon_i \end{aligned} \quad (2)$$

Enrichment analyses

Following established protocols for pathway enrichment analysis of ranked gene lists encompassing a large portion of the genome, we conducted PreRank Gene Set Enrichment Analysis (GSEAPreranked), a threshold-free method that evaluates all measured genes without prior filtering [22]. Initially, we computed ranks for the entire gene list derived from the regression analysis, utilizing the formula $r = -\log_{10}(p\text{-value}) * \text{sign}(\beta)$, which encapsulates both the significance and direction of differential gene expression.

For the enrichment analysis, we employed GSEA version 4.3.2 (v4.3.2) (accessible at <https://www.genepattern.org/modules/docs/GSEAPreranked/1#gsc.tab=0>). Our focus was on five major reference databases sourced from the Molecular Signatures Database (MSigDB v.7.4) [23], comprising two for pathway enrichment (Canonical Pathways including Reactome with 1615 gene sets and KEGG with 186 gene sets) and three for gene ontology (GO Biological Process ontology with 7647 gene sets, GO Cellular Component ontology with 1015 gene sets, and GO Molecular Function ontology with 1799 gene sets). Statistically significant pathways were identified based on adjusted p -values, with a threshold of $p < 0.05$ (two-tailed).

Results

Characteristics of participants with and without muscle biopsies

Table 1 shows the demographics and anthropometric characteristics of participants who underwent muscle biopsies. Consistent with the larger CALERIE enrollment, the subgroup of participants who consented to muscle biopsies was predominantly white and female, with an average age at baseline of 40.1 ± 1.1 years in

the AL group and 38.4 ± 1.0 years in the CR group ($p = 0.249$). From the total participant enrollment for CALERIE [24], biopsy samples were obtained from 93 participants at baseline (43% of all participants), 44 participants (20%) at 12 months, and 31 participants (14%) at 24 months. Among the participants with biopsy data, age and sex were balanced between the AL and CR groups at baseline and participants missing from the 12- and 24-month time points do not differ from those participants without biopsy data in age or group distribution of sex and CR. The characteristics of participants with and without biopsy data did not differ from those who were not biopsied, except for race at 24 months ($p = 0.012$) (Table S1).

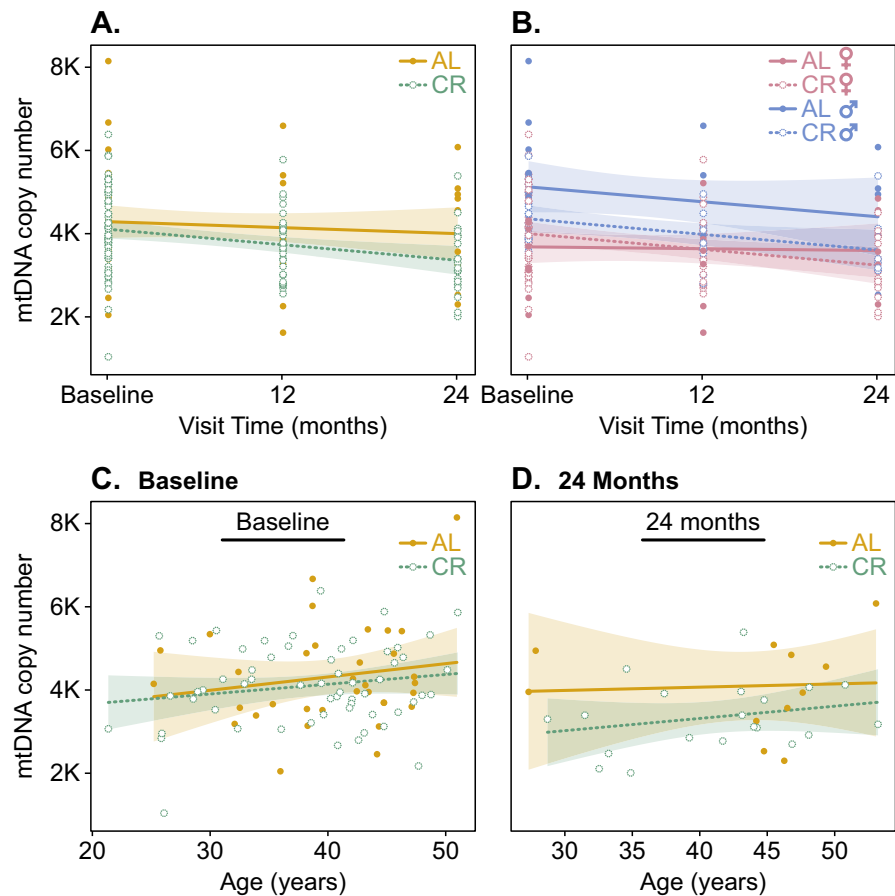
Longitudinal analysis of quadriceps mtDNA copy number

We analyzed mtDNA copy number between AL and CR groups with an LME model. The model included fixed effects of visit time, group, age, and sex. The random intercept was included to account for variability across participants. No statistically significant interaction effects were detected. The analysis revealed effects of time ($\beta = -27.309$, $SE = 6.7850$, $p = 0.0001$) and sex ($\beta = 688.059$, $SE = 190.3941$, $p = 0.0005$) (Fig. 1A and B) on mtDNA copy number. Specifically, mtDNA copy number declined through the 2 years of the study and was lower in females than in males. However, no significant effects of group ($\beta = -198.559$, $SE = 186.4805$, $p = 0.2898$) or age ($\beta = 9.533$, $SE = 12.9616$, $p = 0.4644$, Fig. 1C and D) on mtDNA copy number were observed.

Longitudinal analysis of quadriceps mtDNA deletion mutation frequency

Similarly, we used an LME model to explore the effect of visit time, group, age, and sex on $\log_{10}$ transformed mtDNA deletion mutation frequency. No statistically significant effects of visit time, group, or sex were observed (Fig. 2A and B). However, mtDNA deletion mutation frequency showed a statistically significant increase with age ($p = 0.0041$, Fig. 2C). The difference in the fitted linear regression for the CR group compared to AL at 24 months (Figure S1) is influenced predominantly by two data points.

Fig. 1 mtDNA copy number in CALERIE quadriceps biopsies. **A**, AL versus CR mtDNA copy number per diploid nucleus at baseline, 12, and 24 months. **B**, AL versus CR mtDNA copy number per diploid nucleus by treatment and sex at baseline, 12, and 24 months. Lines are connecting the mean values. mtDNA copy number versus age at baseline (**C**) and after 24 months (**D**). Fitted linear regression lines with 95% confidence bands



MtDNA copy number and mutation frequency predict multiple muscle-related physiological measures

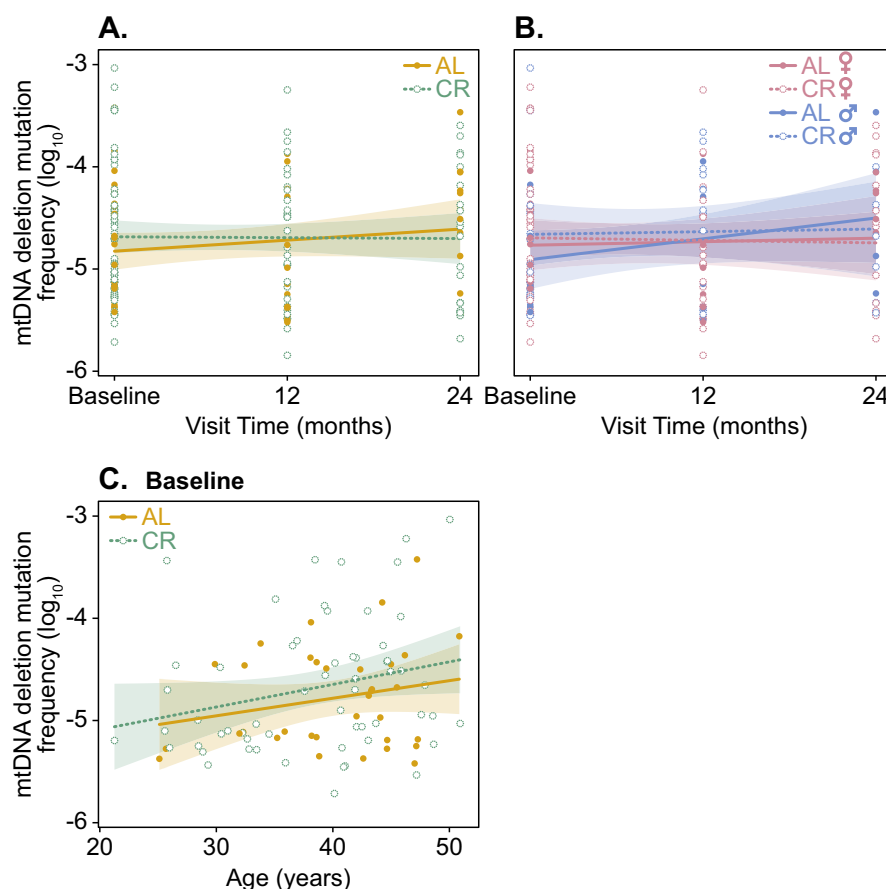
We previously observed that mtDNA copy number and deletion mutation frequency correlates with several physiologic measures in older adults [12]. In CALERIE, we examined the correlations between mtDNA copy number/deletion frequency and measures of muscle mass and whole-body metabolism (Fig. 3). This analysis was limited to the baseline dataset given the low number of participants who consented to biopsies at later visits. MtDNA copy number showed weak to moderate positive correlations to whole-body fat-free mass ($r=0.43$, $p=2e-05$), KCAL ($r=0.34$, $p=0.00077$), RMR ($r=0.42$, $p=0.00261$), TEERQ ($r=0.31$, $p=0.00261$) and $VO_2\text{max}$ ($r=0.41$, $p=1e-04$). Of these measures, mtDNA deletion mutation frequency showed a statistically significant but relatively weak

negative correlation with $VO_2\text{max}$ (mL/kg/min) ($r=-0.23$, $p=3e-05$).

Associations between mtDNA copy number and gene expression at baseline in CALERIE quadriceps biopsies

First, we examined correlations of mtDNA copy number to gene expression using a linear regression model adjusted for age, sex, race, and technical batch in the skeletal muscle biopsies at baseline (Fig. 4). The expression of 606 genes was correlated with mtDNA copy number at baseline ($p<0.01$, Figs. 4A, S6). Examining the 30 genes with the greatest absolute beta values (i.e., the strongest association with mtDNA copy number), we found effect sizes ranging from medium to large (Fig. 4B). All these genes exhibit a clear association with mtDNA copy number at baseline

Fig. 2 mtDNA deletion mutation frequency in CALERIE quadriceps biopsies. **A.** AL versus CR mtDNA deletion mutation frequency at baseline, 12, and 24 months. **B.** AL versus CR mtDNA deletion mutation frequency by sex at baseline, 12, and 24 months. Lines are connecting the mean values at each time point. **C.** AL versus CR mtDNA deletion mutation frequency versus age at baseline



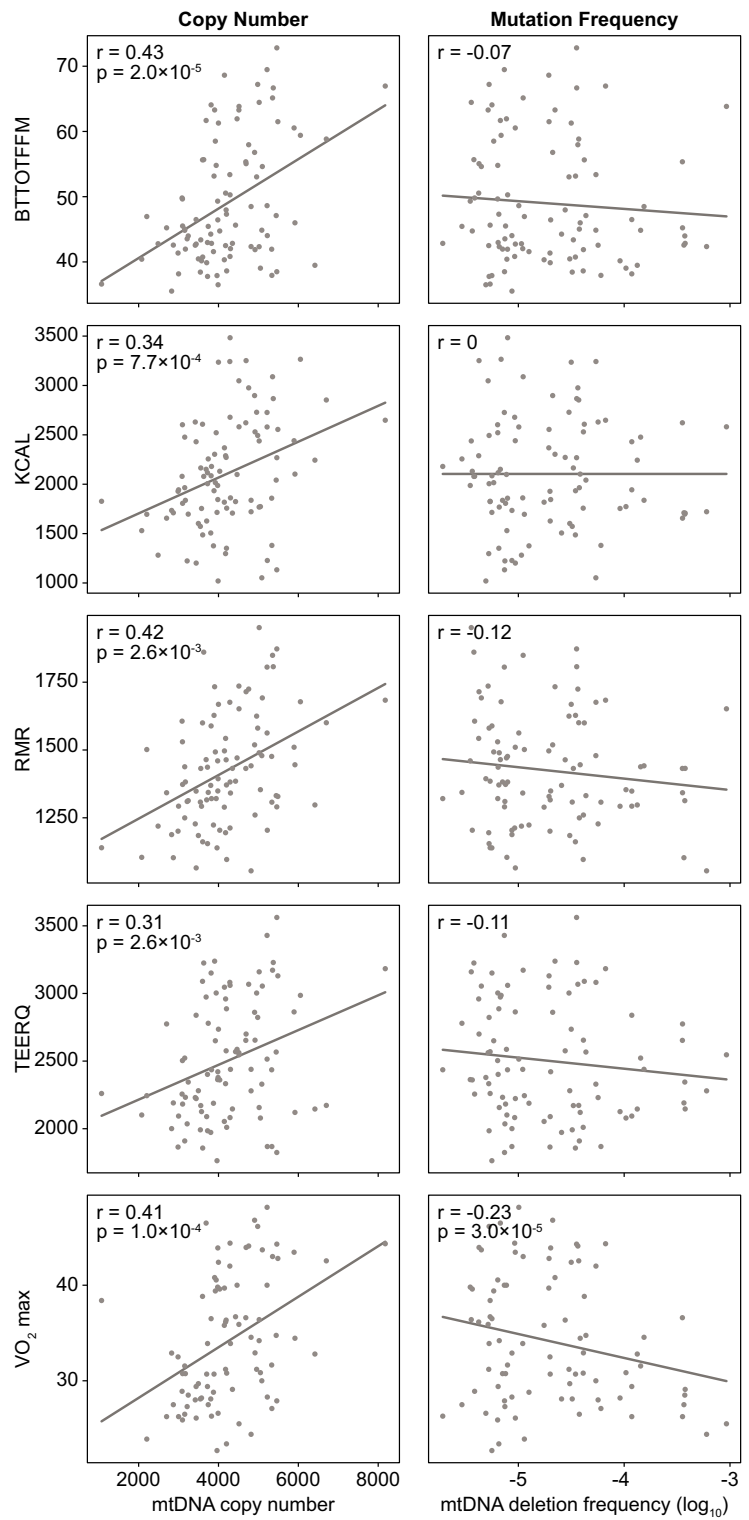
(Fig. 4C and 4D). Most of these 30 genes were not directly related to mitochondrial function except for MTRNR2L12, HMOX1, and NME9. However, pre-Rank Gene Set Enrichment Analysis (see Materials and methods) taking the full list of genes showed overlapping gene ontologies and pathways that pointed to transmembrane electron transport, oxidative phosphorylation, and mitochondrial function as well as neurodegenerative diseases (Fig. 5). Down-regulated gene ontologies and pathways at baseline were enriched for nuclear genome processes, splicing, and ribosomes.

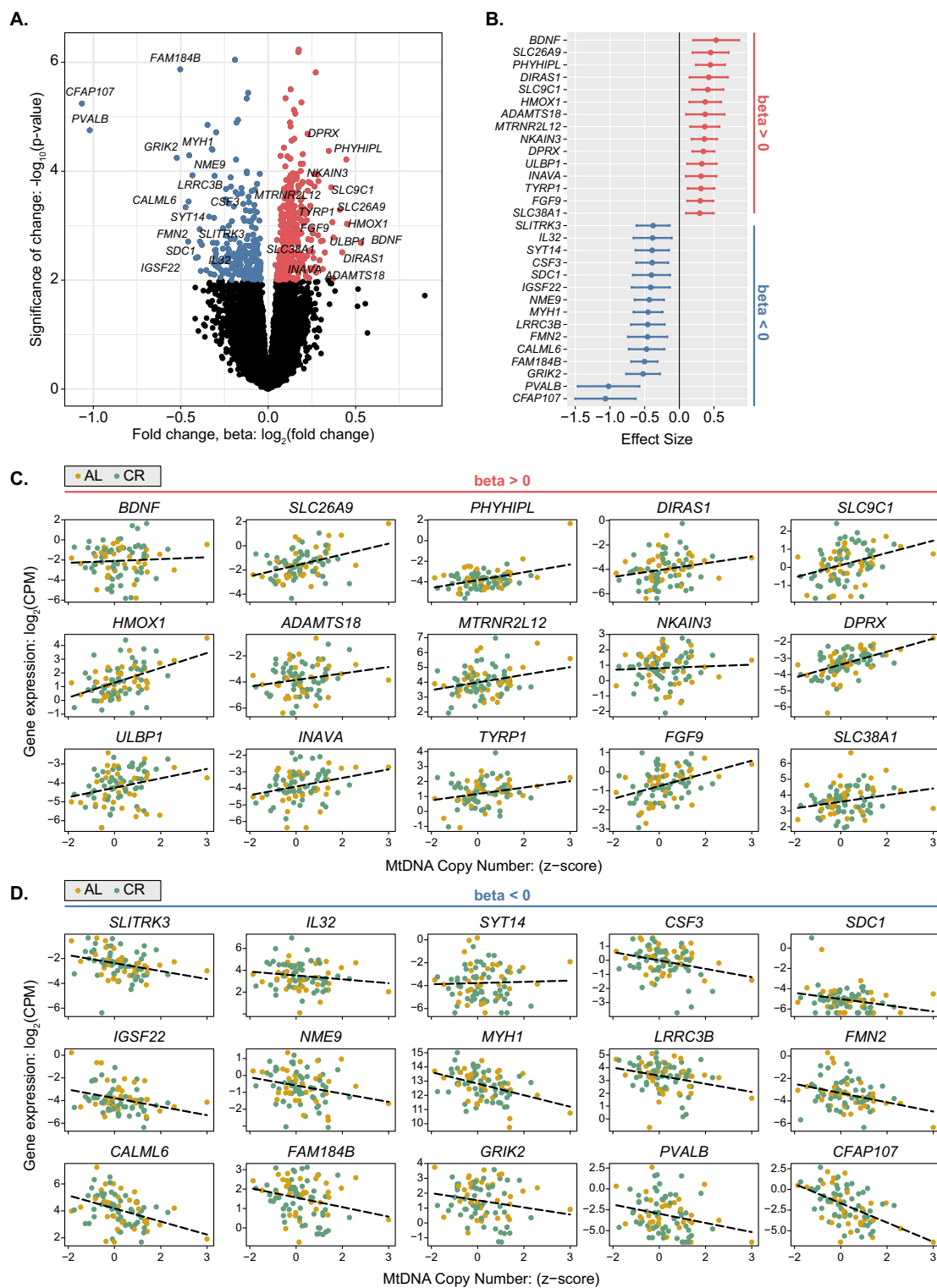
Associations between mtDNA copy number and gene expression in longitudinal CALERIE muscle biopsies

Next, we examined associations between gene expression and mtDNA copy number using a linear mixed-effect model through the interaction among group (CR vs. AL) and linear time (year) adjusted

for age, sex, race, and technical batch (Methods). The number of genes whose expression was associated with mtDNA copy number declined to 286 (from 606 at baseline) and only six of the baseline genes correlated with mtDNA copy number at the later time points (Figs. 6A, S6). One gene whose expression correlated with mtDNA copy number at baseline and later visits was MTRNR2L12, a member of the humanin family of mitochondrial-derived peptides that have various protective roles, including in mitochondria [25]. Examining the 30 genes with the greatest absolute beta values, effect sizes again ranged from medium to large (Fig. 6B). Two of these 30 genes are directly associated with mitochondria—MTRNR2L12, mentioned above, and MTRNR2L8, while the remainder have no clear role in mitochondria. In general, the changes in repeated measures seen between mtDNA copy number and gene expression in the AL samples were blunted by CR in the longitudinal analysis (Fig. 6C

Fig. 3 mtDNA copy number predicts multiple physiologic measures. Fitted linear regression lines





◀**Fig. 4** mtDNA copy number association with gene expression at baseline. **A** Volcano plot depicting differential changes of gene expression using LMM and mtDNA copy number. Red dot (positive beta: $\beta+$) represents genes whose differential expression is significantly increased with mtDNA copy number; blue dot (for negative beta: $\beta-$) represents genes whose expression is significantly decreased with mtDNA copy number. The top 15 genes with the largest β (positive or negative) in each group were labeled. **B** Forest plot of top 40 transcripts showing range of β coefficient (15 with $\beta+$ and 15 with $\beta-$) and standard errors. **C** Linear regression plots illustrate the relationship between mtDNA copy number and gene expression for the top 15 genes with the largest β (positive or negative) at baseline. In this analysis, gene expression serves as the outcome variable, while mtDNA copy number acts as the predictor, adjusted for age, sex, and race. The upper two rows depict genes positively associated, while the bottom two rows show negatively associated genes. Genes associated with subjects who were randomized to caloric restriction (CR) are highlighted in green, while those randomized to ad libitum (AL) are in yellow. The x-axis represents mtDNA copy number, and the y-axis represents RNA expression ($\log_2(\text{CPM})$) at baseline

and D). The upregulated gene ontologies and pathways differed substantially from baseline with an emphasis on extracellular matrix, membrane proteins, and immune cell activation (Fig. 7). Downregulated genes shifted to mitochondrial and protein translation.

Associations between mtDNA deletion mutation frequency and gene expression at baseline in CALERIE muscle biopsies

For mtDNA deletion mutation frequency at baseline, we found 362 genes whose expression was associated with mutation frequency ($p < 0.01$, Figure S2A); eight of those were also associated with mtDNA copy number at baseline (Figure S6). Examining the 30 genes with the greatest absolute beta values, we found effect sizes that were generally smaller than those noted for mtDNA copy number correlations (Figure S2B). Among these genes, most again are not directly related to mitochondrial function except for AIFM3, which is involved in apoptosis and mitochondrial membrane potential [26]. Gene ontology and pathway analysis for genes associated with mtDNA deletion frequency at baseline differed largely from those observed with mtDNA copy number (Figure S3). Upregulated ontologies and pathways were primarily related to inflammation. Downregulated categories were primarily protein translation and mitochondrial, which was similar to the analysis at 12 months.

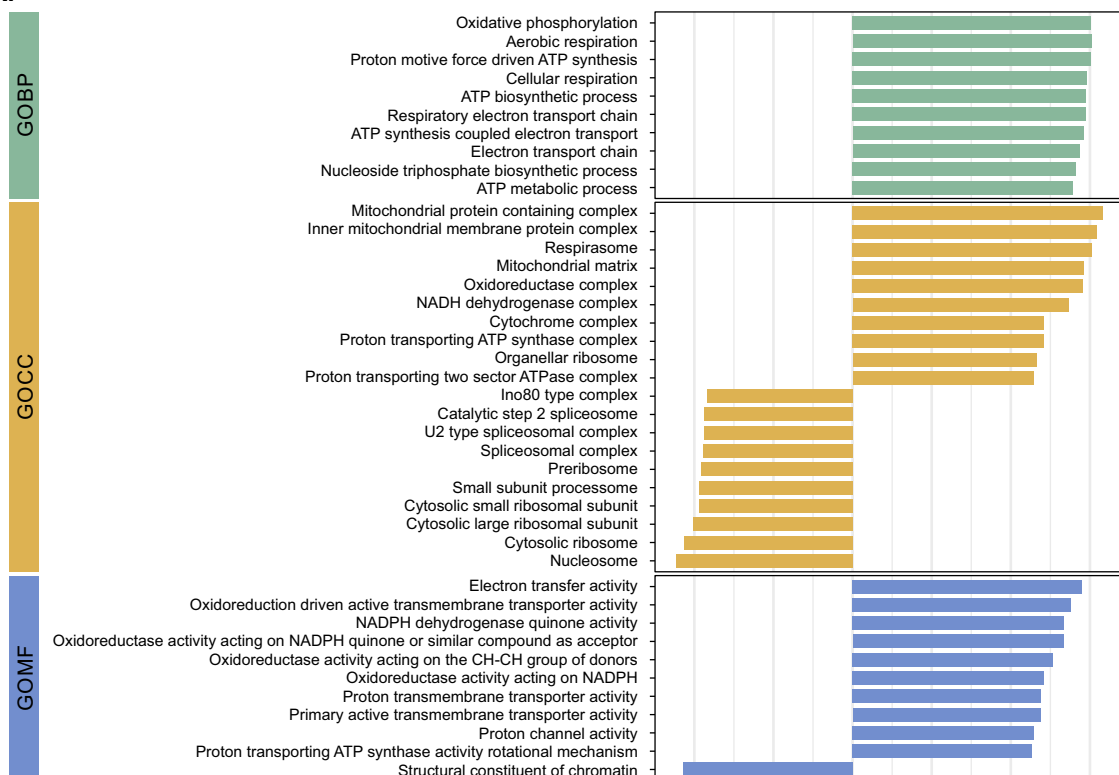
Associations between mtDNA deletion mutation frequency and gene expression in longitudinal CALERIE muscle biopsies

Lastly, we used linear mixed-effect models to examine whether mtDNA mutations were differentially associated with gene expression changes with time according to group, namely CR and AL, adjusting for age, sex, race, and technical batch. The number of genes whose expression was associated with mtDNA mutation frequency in the longitudinal analysis declined to 212 and only 2 genes correlated with mtDNA mutation frequency in the baseline analysis (Figures S4A, S6). Examining the 30 genes with the greatest absolute beta values, effect sizes again ranged from medium to large (Figure S4B). One gene whose expression correlated with mtDNA mutation frequency in the longitudinal analysis was IRGM (Immunity-related GTPase family M protein), which is involved in autophagy [27, 28]. As seen with mtDNA copy number, CR often blunted the changes in repeated measures seen between mtDNA mutation frequency and gene expression as compared to AL samples (Figure S4C-D). The gene ontologies and pathways were generally upregulated in the *cellular respiration* and *mitochondrial* categories and downregulated in the *resolution of D-loop structures* category (Figure S5).

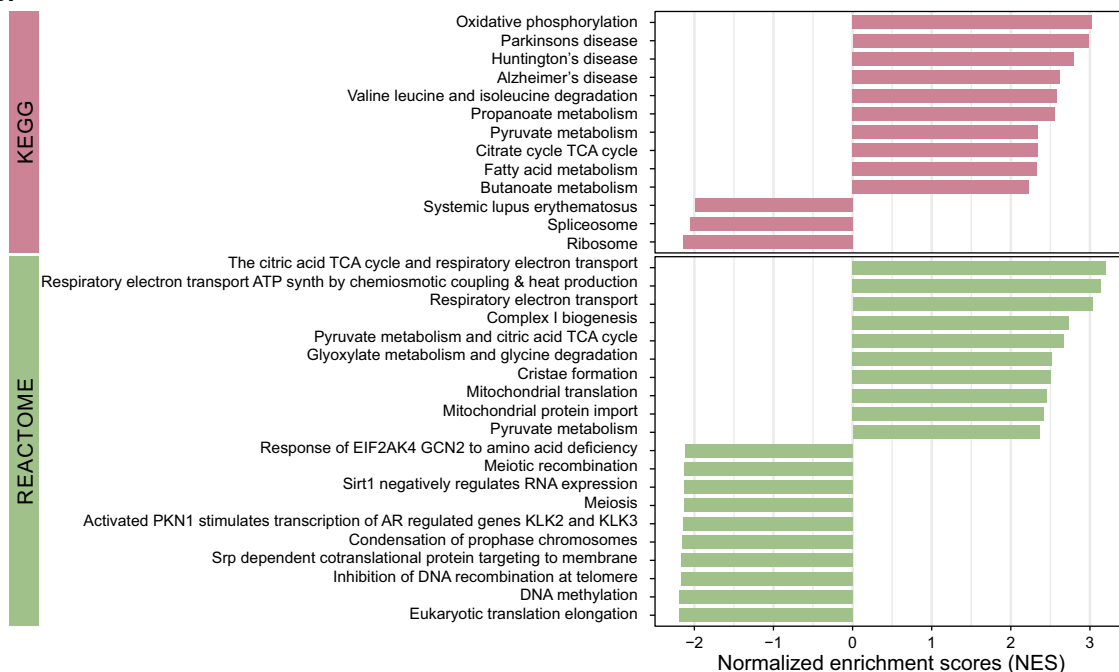
Discussion

The present study yielded several novel findings regarding the mitochondrial genome in human quadriceps muscle during a trial of CR. First, sex and time affect mtDNA copy number, with lower copy number in females and a decline in copy number over the 2-year intervention or control period. Secondly, on average, the observed mtDNA deletion mutation frequency was lower with CR and differed between women and men, but these differences were not statistically significant, likely because our sample size was too small. Our final novel finding was that mtDNA copy number and mutation frequency correlate with the expression of numerous genes, some of which suggest enrichment for oxidative phosphorylation and mitochondrial function, indicating signaling between the mitochondrial and nuclear genomes. Confirmatory findings in this study include the observation that, even in these relatively young participants,

A.



B.



◀Fig. 5 Significantly enriched gene ontology terms and pathways (adjusted p -value < 0.05) associated with mtDNA copy number at baseline study. Enrichment analysis is performed using pre-ranked gene set enrichment analysis (GSEAPre-ranked) where genes were ranked based on a combination of p -values and the sign of beta coefficients derived from linear associations between mtDNA copy number and gene expression at baseline (Methods). The top panel displays enriched gene ontology terms from Molecular Function (MF), Cellular Component (CC), and Biological Process (BP) categories, while the bottom panel shows enriched functional domain pathways from Reactome (R) and KEGG (K) databases. The right column indicates the net enrichment score (NES) for each gene ontology or pathway, with upregulated terms shown in orange (NES > 0) and downregulated terms in blue (NES < 0) with respect to mtDNA copy number. The top 10 most significant gene ontology terms and pathways (based on absolute NES score) are presented from each category

older age was associated with higher quadriceps mtDNA deletion mutation frequency as we have previously observed [11, 13]. We also again observed that mtDNA copy number and deletion mutation frequency predicted some measures that relate to skeletal muscle physiology and metabolism [29].

The effects of CR on quadriceps mtDNA copy number have been explored in two prior human trials with conflicting results. A 6-month study reported a 35% increase in mtDNA copy number and evidence of upregulated mitochondrial biogenesis [9], while a 12-month study found a ~20% decrease in mtDNA copy number in both CR and control groups, with no evidence of biogenesis upregulation [8]. Our analysis of CALERIE muscle biopsy samples found no significant effect of CR on skeletal muscle mtDNA copy number. Differences across studies may stem from the duration of CR, the level of CR, or methodological variations in measuring mtDNA copy number. The higher mtDNA copy number observed in male compared to female quadriceps muscle aligns with previous findings in older individuals [12]. Regarding mtDNA deletion mutation frequency, we observed an age-related increase in the human quadriceps, consistent with prior studies in mice, rats, rhesus monkeys, and other human cohorts [12, 30–34]. These human measurements, obtained from blinded samples, support the observation that mtDNA deletions increase with aging.

The conflicting reports of mtDNA copy number in CR studies highlight the need for standardized measurement methods. While standards for nuclear genome copy number variation exist [35],

similar protocols for mtDNA are lacking. Normalizing mtDNA copy number to nuclear DNA (nDNA) is common, yet this approach may be influenced by age-related changes in nuclear content, such as inflammatory infiltrates or cellular polyploidy. Further variation arises from differences in nDNA targets used for normalization, with some not being single-copy genes or failing to account for nuclear diploidy. These methodological inconsistencies limit direct comparisons across studies and preclude combining our findings with other human CR data.

Age is increasingly recognized as the primary risk factor for the leading causes of death in the USA [36]. Life-extending interventions such as CR are poised for clinical translation; however, survival is not a practical endpoint. Measures are needed to determine the rate of biological aging and predict healthspan and lifespan [37]. Such metrics should aid in predicting the progression of age-related phenotypes, selecting interventions, and evaluating outcomes of interventions [38]. Mitochondrial bioenergetics are correlated with longevity and are attractive targets for drug development and therapeutic trials (e.g., CR, metformin, exercise) [9, 39–42]. As such, metrics of mitochondrial fitness may be particularly well suited as biomarkers. Indeed, studies have consistently demonstrated negative correlations between interventions that alter lifespan and mtDNA deletion frequency/ETC deficient cell abundance [43–45].

The duration of CR in CALERIE was relatively short (i.e., 2 years) in the context of human lifespan and the pace of age-induced physiological changes. Longer duration human CR trials will be needed to elucidate the age-specific effects of CR, including potential adverse effects, as well as subtle age-induced changes that emerge over longer time frames. For example, given the increased rate of mtDNA mutation frequency with time observed in the current study, a CR study of 4 years is more likely to uncover effects of CR on human muscle mtDNA deletion frequency, as measured in mice, rats, and non-human primates [43, 46–49].

The CALERIE participants were healthy adults without obesity, which limits the generalizability of the findings to other populations, such as those with chronic conditions, older adults, or individuals with higher BMI. The stringent eligibility criteria may mask the variability of outcomes that are likely in a more diverse participant population. Across inbred

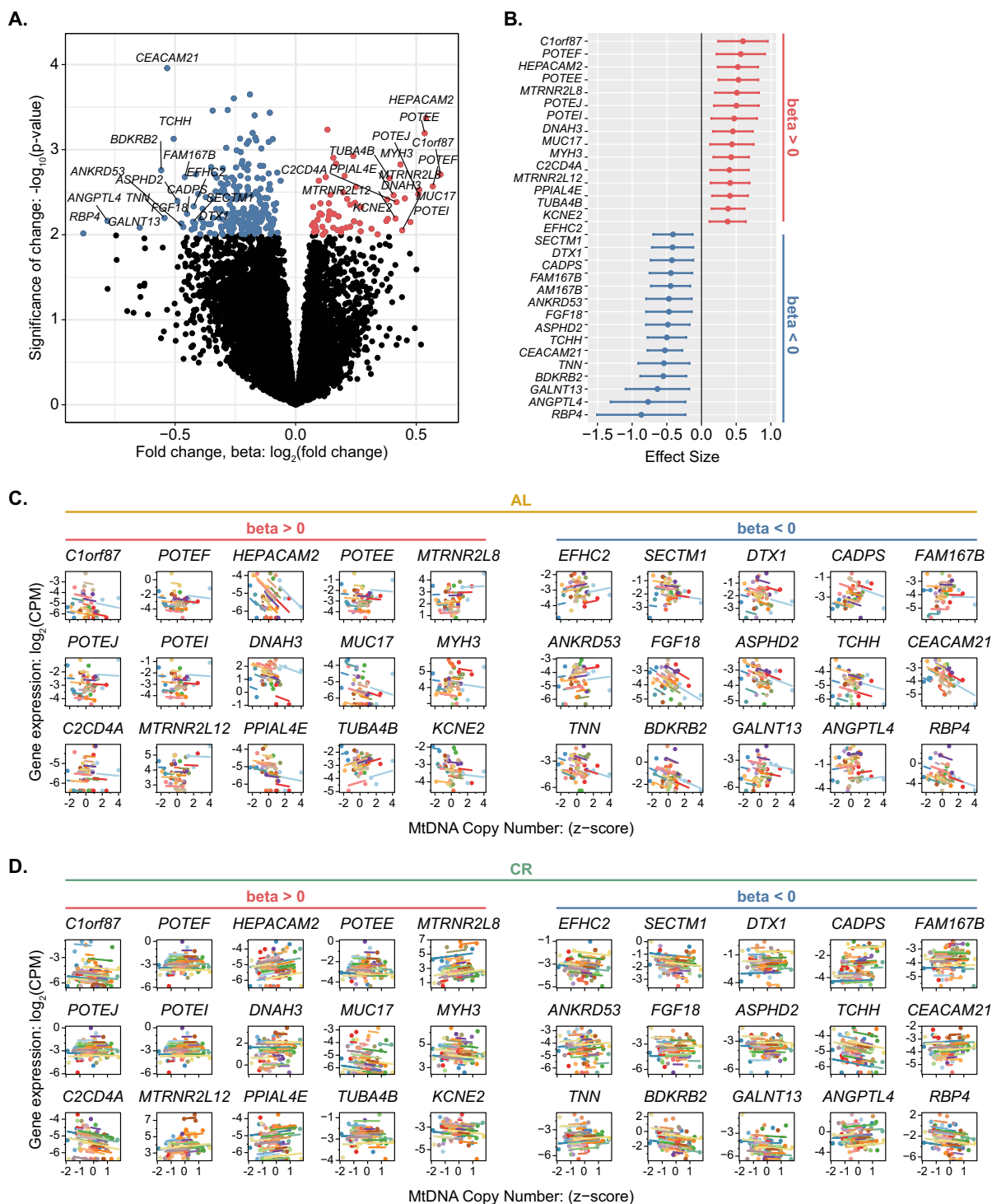


Fig. 6 Longitudinal analysis of mtDNA copy number association with gene expression. **A.** Volcano plot depicting differential changes of gene expression using LMM and mtDNA copy number. Red dot (positive beta: $\beta+$) represents genes whose differential expression is significantly increased with mtDNA copy number; blue dot (for negative beta: $\beta-$) represents genes whose expression is significantly decreased with mtDNA copy number. The top 15 genes with the largest β (positive or negative) in each group were labeled. **B.** Forest plot of top 40 transcripts showing range of β coefficient (15 with $\beta+$ and 15 with $\beta-$) and standard errors. **C.** Repetitive measure plot showing linear trend of each individual subject (sample with at least two time point data) for treatment CR group (right) and AL group (left) separately for topmost 30 significant genes as shown in A and B. In the plot, x-axis represents mtDNA copy number (z-score) and y-axis represents gene expression value ($\log_2(\text{CPM})$). Lines and data points from the same subject have the same color. The first two rows both for CR and AL groups are genes with beta coefficient positive in treatment CR group with reference to AL group, whereas last two represents genes with beta coefficient negative in treatment CR group with reference to AL group. Genes are labeled according to their respective gene symbols

rodent strains with differing genetic backgrounds, CR is known to have differential effects on lifespan and skeletal muscle [50, 51], suggesting that there will be similar variability when CR is deployed across a diverse human population. Furthermore, the diversity of the mitochondrial genome closely parallels human diversity [52] and changes in mitochondrial DNA integrity involve the interplay of nuclear DNA and mitochondrial DNA variants [53]. Future human CR studies with more diverse participants may help to begin uncovering these complex dynamics.

A limitation of our study is the lack of statistical power for some of our comparisons, particularly at later time points in the trial. The decline in statistical power was due to either attrition or non-agreement to have subsequent biopsies as the trial progressed (Table 1). The effect of the CR intervention on mtDNA deletion mutation frequency may become more apparent over an extended intervention period. In addition to a longer duration of CR, future studies of the mitochondrial genome in human CR trials will benefit from higher participant consent for muscle biopsies. Improvement of patient comfort through the use of smaller gauge biopsy needles [54] may be one approach to improve participant recruitment and retention in future studies. The use of smaller biopsy needles is compatible with our mtDNA assays that

currently require less than 10 mg of tissue or one microgram of total DNA.

The CALERIE trial has provided insights into the potential benefits and mechanisms of caloric restriction in humans [19, 20, 55–60]. The present study adds value by leveraging skeletal muscle biopsy samples from CALERIE and adding metrics of mtDNA integrity and their relationships to gene expression to extend the CALERIE Genomic Data Resource (PMID: 39229162). Future studies will build on the foundational knowledge provided by the CALERIE trial, ultimately contributing to more effective interventions for promoting healthy aging and longevity.

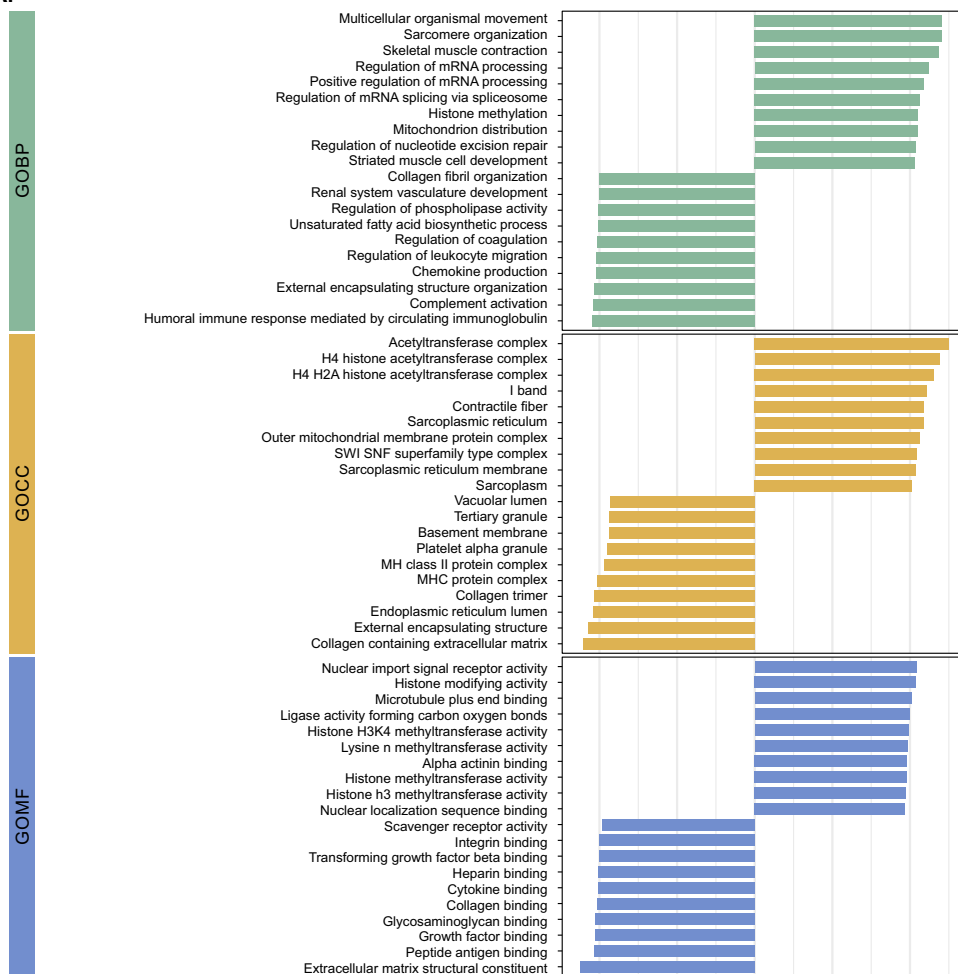
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Author contribution Conceptualization, methodology, supervision, and funding acquisition for the CALERIE trial included contributions from Leanne M. Redman, Corby K. Martin, Sai Krupa Das, Susan B. Racette, William E. Kraus, and Kim M. Huffman. For the mtDNA studies, material preparation and data collection were performed by Allen Herbst, Debbie McKenzie, Judd M. Aiken, Steve Horvath, and Jonathan Wanagat. Data analysis was performed by Jayanta K. Das, Yu Zhang, Allen Herbst, Nianjun Liu, David B. Allison, Steve Horvath, Julian Candia, Nirad Banskota, Luigi Ferrucci, and Jonathan Wanagat. The first draft of the manuscript was written by Jayanta K. Das, Yu Zhang, Allen Herbst, and Jonathan Wanagat, and all authors commented on subsequent versions of the manuscript. All authors read and approved the final manuscript.

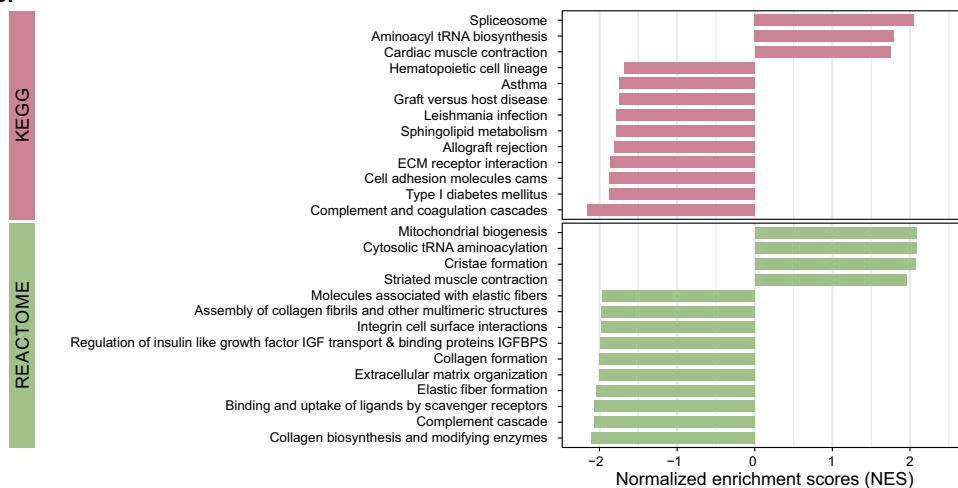
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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

A.



B.



Normalized enrichment scores (NES)

◀**Fig. 7** Significantly enriched gene ontology terms and pathways (adjusted p -value < 0.05) associated with mtDNA copy number in longitudinal study. Enrichment analysis is performed using preranked gene set enrichment analysis (GSEAPreranked) where genes were ranked based on a combination of p -values and the sign of beta coefficients derived from linear associations between mtDNA copy number and gene expression in longitudinal study (Methods). The top panel displays enriched gene ontology terms from Molecular Function (MF), Cellular Component (CC), and Biological Process (BP) categories, while the bottom panel shows enriched functional domain pathways from Reactome (R) and KEGG (K) databases. The right column indicates the net enrichment score (NES) for each gene ontology or pathway, with upregulated terms shown in orange (NES > 0) and downregulated terms in blue (NES < 0) with respect to mtDNA copy number. The top 10 most significant gene ontology terms and pathways (based on absolute NES score) are presented from each category

Declarations

Ethics approval and consent to participate Not applicable as this ancillary study was performed on de-identified samples and did not include human subjects.

Conflict of interest The authors have no conflicts of interest to declare. Steve Horvath works for Altos Labs, a biotechnology company. Dr. Allison and his institutions have received payments, grants, in-kind donations, contracts, and travel expenses from many for-profit and nonprofit organizations, including food, pharmaceutical, dietary supplement, and litigating firms, but none directly related to this project.

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References

- Jaul E, Barron J. Age-related diseases and clinical and public health implications for the 85 years old and over population. *Front Public Health*. 2017;5:335.
- Amorim JA, et al. Mitochondrial and metabolic dysfunction in ageing and age-related diseases. *Nat Rev Endocrinol*. 2022;18(4):243–58.
- Picca A, et al. The contribution of mitochondrial DNA alterations to aging, cancer, and neurodegeneration. *Exp Gerontol*. 2023;178:112203.
- Wiesner RJ, Zsurka G, Kunz WS. Mitochondrial DNA damage and the aging process: facts and imaginations. *Free Radic Res*. 2006;40(12):1284–94.
- Lopez-Lluch G, Navas P. Calorie restriction as an intervention in ageing. *J Physiol*. 2016;594(8):2043–60.
- Most J, et al. Calorie restriction in humans: an update. *Ageing Res Rev*. 2017;39:36–45.
- Lanza IR, et al. Chronic caloric restriction preserves mitochondrial function in senescence without increasing mitochondrial biogenesis. *Cell Metab*. 2012;16(6):777–88.
- Sparks LM, et al. Effects of 12 months of caloric restriction on muscle mitochondrial function in healthy individuals. *J Clin Endocrinol Metab*. 2017;102(1):111–21.
- Civitarese AE, et al. Calorie restriction increases muscle mitochondrial biogenesis in healthy humans. *PLoS Med*. 2007;4(3):e76.
- Green CL, Lamm DW, Fontana L. Molecular mechanisms of dietary restriction promoting health and longevity. *Nat Rev Mol Cell Biol*. 2022;23(1):56–73.
- Herbst A, et al. Mitochondrial DNA deletion mutations increase exponentially with age in human skeletal muscle. *Aging Clin Exp Res*. 2021;33(7):1811–20.
- Herbst A, et al. Skeletal muscle mitochondrial DNA copy number and mitochondrial DNA deletion mutation frequency as predictors of physical performance in older men and women. *Geroscience*. 2021;43(3):1253–64.
- Vandiver AR, et al. Nanopore sequencing identifies a higher frequency and expanded spectrum of mitochondrial DNA deletion mutations in human aging. *Aging Cell*. 2023;22(6):e13842.
- Herbst A, et al. Latent mitochondrial DNA deletion mutations drive muscle fiber loss at old age. *Aging Cell*. 2016;15(6):1132–9.
- Wanagat J, et al. Mitochondrial DNA deletion mutations colocalize with segmental electron transport system abnormalities, muscle fiber atrophy, fiber splitting, and oxidative damage in sarcopenia. *FASEB J*. 2001;15(2):322–32.
- Zhang X, et al. The distinct spatiotemporal distribution and effect of feed restriction on mtDNA copy number in broilers. *Sci Rep*. 2020;10(1):3240. <https://doi.org/10.1038/s41598-020-60123-1>
- Coccaro N, et al. Digital PCR: A reliable tool for analyzing and monitoring hematologic malignancies. *Int J Mol Sci*. 2020. <https://doi.org/10.3390/ijms21093141>.
- Redman LM, Ravussin E. Caloric restriction in humans: impact on physiological, psychological, and behavioral outcomes. *Antioxid Redox Signal*. 2011;14(2):275–87.
- Ravussin E, et al. A 2-year randomized controlled trial of human caloric restriction: feasibility and effects on predictors of health span and longevity. *J Gerontol A Biol Sci Med Sci*. 2015;70(9):1097–104.
- Das JK, et al. Calorie restriction modulates the transcription of genes related to stress response and longevity in human muscle: the CALERIE study. *Aging Cell*. 2023;22(12):e13963.

21. Twisk J, et al. Multiple imputation of missing values was not necessary before performing a longitudinal mixed-model analysis. *J Clin Epidemiol*. 2013;66(9):1022–8.
22. Reimand J, et al. Pathway enrichment analysis and visualization of omics data using g:Profiler, GSEA, Cytoscape and EnrichmentMap. *Nat Protoc*. 2019;14(2):482–517.
23. Subramanian A, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A*. 2005;102(43):15545–50.
24. Bales CW, Kraus WE. Caloric restriction: implications for human cardiometabolic health. *J Cardiopulm Rehabil Prev*. 2013;33(4):201–8.
25. Guo B, et al. Humanin peptide suppresses apoptosis by interfering with Bax activation. *Nature*. 2003;423(6938):456–61.
26. Xie Q, et al. Molecular cloning and characterization of a human AIF-like gene with ability to induce apoptosis. *J Biol Chem*. 2005;280(20):19673–81.
27. Chauhan S, Mandell MA, Deretic V. IRGM governs the core autophagy machinery to conduct antimicrobial defense. *Mol Cell*. 2015;58(3):507–21.
28. Singh SB, et al. Human IRGM induces autophagy to eliminate intracellular mycobacteria. *Science*. 2006;313(5792):1438–41.
29. Herbst A, et al. Mitochondrial DNA deletion mutations increase exponentially with age in human skeletal muscle. *Aging Clin Exp Res*. 2020. <https://doi.org/10.1007/s40520-020-01698-7>.
30. Aspnes LE, et al. Caloric restriction reduces fiber loss and mitochondrial abnormalities in aged rat muscle. *FASEB J*. 1997;11(7):573–81.
31. Bua E, et al. Mitochondrial DNA-deletion mutations accumulate intracellularly to detrimental levels in aged human skeletal muscle fibers. *Am J Hum Genet*. 2006;79(3):469–80.
32. Chung SS, et al. Analysis of age-associated mitochondrial DNA deletion breakpoint regions from mice suggests a novel model of deletion formation. *Age*. 1996;19:117–28.
33. Herbst A, et al. Age- and time-dependent mitochondrial genotoxic and myopathic effects of beta-guanidinopropionic acid, a creatine analog, on rodent skeletal muscles. *GeroScience*. 2023;45(1):555–67.
34. Lee CM, et al. Multiple mitochondrial DNA deletions associated with age in skeletal muscle of rhesus monkeys. *J Gerontol*. 1993;48(6):B201–5.
35. Riggs ER, et al. Technical standards for the interpretation and reporting of constitutional copy-number variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics (ACMG) and the Clinical Genome Resource (ClinGen). *Genet Med*. 2020;22(2):245–57.
36. Sierra F. The emergence of geroscience as an interdisciplinary approach to the enhancement of health span and life span. *Cold Spring Harb Perspect Med*. 2016;6(4):a025163.
37. Ferrucci L, et al. Measuring biological aging in humans: a quest. *Aging Cell*. 2020;19(2):e13080.
38. Moqri M, Herzog C, Poganik JR, Ying K, Justice JN, Belsky DW, Higgins-Chen AT, Chen BH, Cohen AA, Fuellen G, Hägg S, Marioni RE, Widschwendter M, Fortney K, Fedichev PO, Zhavoronkov A, Barzilai N, Lasky-Su J, Kiel DP, Kennedy BK, Cummings S, Slagboom PE, Verdin E, Maier AB, Sebastiano V, Snyder MP, Gladyshev VN, Horvath S, Ferrucci L. Validation of biomarkers of aging. *Nat Med*. 2024 Feb;30(2):360–372.
39. Nisoli E, et al. Calorie restriction promotes mitochondrial biogenesis by inducing the expression of eNOS. *Science*. 2005;310(5746):314–7.
40. Martin-Montalvo A, et al. Metformin improves healthspan and lifespan in mice. *Nat Commun*. 2013;4:2192. <https://doi.org/10.1038/ncomms3192>
41. Lopez-Lluch G, et al. Mitochondrial responsibility in ageing process: innocent, suspect or guilty. *Biogerontology*. 2015;16(5):599–620.
42. Longo VD, et al. Interventions to slow aging in humans: are we ready? *Aging Cell*. 2015;14(4):497–510.
43. McKiernan SH, et al. Adult-onset calorie restriction delays the accumulation of mitochondrial enzyme abnormalities in aging rat kidney tubular epithelial cells. *Am J Physiol Renal Physiol*. 2007;292(6):F1751–60.
44. Dai DF, et al. Age-dependent cardiomyopathy in mitochondrial mutator mice is attenuated by overexpression of catalase targeted to mitochondria. *Aging Cell*. 2010;9(4):536–44.
45. Bielas J, et al. Long term rapamycin treatment improves mitochondrial DNA quality in aging mice. *Exp Gerontol*. 2018;106:125–31.
46. Bua E, McKiernan SH, Aiken JM. Calorie restriction limits the generation but not the progression of mitochondrial abnormalities in aging skeletal muscle. *FASEB J*. 2004;18(3):582–4.
47. McKiernan SH, et al. Early-onset calorie restriction conserves fiber number in aging rat skeletal muscle. *FASEB J*. 2004;18(3):580–1.
48. McKiernan SH, et al. Cellular adaptation contributes to calorie restriction-induced preservation of skeletal muscle in aged rhesus monkeys. *Exp Gerontol*. 2012;47(3):229–36.
49. McKiernan SH, et al. Calorie restriction delays aging-induced cellular phenotypes in rhesus monkey skeletal muscle. *Exp Gerontol*. 2011;46(1):23–9.
50. Liao CY, et al. Genetic variation in the murine lifespan response to dietary restriction: from life extension to life shortening. *Aging Cell*. 2010;9(1):92–5.
51. Boldrin L, et al. The effect of calorie restriction on mouse skeletal muscle is sex, strain and time-dependent. *Sci Rep*. 2017;7(1):5160. <https://doi.org/10.1038/s41598-017-04896-y>
52. Hernandez CL. Mitochondrial DNA in human diversity and health: from the golden age to the omics era. *Genes*. 2023. <https://doi.org/10.3390/genes14081534>.
53. Koller A, et al. Nuclear and mitochondrial genetic variants associated with mitochondrial DNA copy number. *Sci Rep*. 2024;14(1):2083. <https://doi.org/10.1038/s41598-024-52373-0>

54. Barthelemy F, et al. A well-tolerated core needle muscle biopsy process suitable for children and adults. *Muscle Nerve*. 2020;62(6):688–98.
55. Hastings WJ, et al. Effect of long-term caloric restriction on telomere length in healthy adults: CALERIE 2 trial analysis. *Aging Cell*. 2024;23(6):e14149.
56. Aversa Z, et al. Calorie restriction reduces biomarkers of cellular senescence in humans. *Aging Cell*. 2024;23(2):e14038.
57. Waziry R, et al. Effect of long-term caloric restriction on DNA methylation measures of biological aging in healthy adults from the CALERIE trial. *Nat Aging*. 2023;3(3):248–57.
58. Kraus WE, et al. 2 years of calorie restriction and cardio-metabolic risk (CALERIE): exploratory outcomes of a multicentre, phase 2, randomised controlled trial. *Lancet Diabetes Endocrinol*. 2019;7(9):673–83.
59. Das SK, et al. Body-composition changes in the Comprehensive Assessment of Long-term Effects of Reducing Intake of Energy (CALERIE)-2 study: a 2-y randomized controlled trial of calorie restriction in nonobese humans. *Am J Clin Nutr*. 2017;105(4):913–27.
60. Racette SB, et al. Effects of two years of calorie restriction on aerobic capacity and muscle strength. *Med Sci Sports Exerc*. 2017;49(11):2240–9.

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